RELATIVE PSEUDO-COHERENCE SURVIVES LOCALIZING THE ALGEBRA AND THE BASE RING AT ONCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 21498-21505 (canonical tag 0679) and the counted proof is the proof environment at source lines 21507-21540, which immediately follows it and is its own proof.

Editorial note. The printed text begins a sentence with a lower case write at proof line 21531 and writes A_f for a localization where A_g is meant at proof lines 21527-21529; both are copied exactly as printed and are not corrected.

Editorial note. The preamble span runs to line 90, the current end of the retained preamble, which includes the appended environment definitions; the counted text does not use them, and the span is taken whole so that the file version is unambiguous.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key and by this note. The counted statement and proof cite no bibliography entry, so this card needs no bibliography.

Allowed standard prerequisites (editorial).

- A complex that is m-pseudo-coherent relative to a base remains so when it is viewed as a complex over a larger polynomial algebra over the same base, provided the algebra used is pseudo-coherent as a module over that larger algebra. invoked at proof lines 21515-21518 and again at proof lines 21535-21539. Moves relative pseudo-coherence from the localized base to a polynomial ring over the base and then to the quotient presenting the further localization.
- The localization of a ring at an element is the quotient of a polynomial ring by the relation expressing the inverse of that element, and the same holds for a second localization. used at proof lines 21509-21511 and 21531-21534. Provides the explicit presentations in terms of which the two moves are made.
- Pulling back a relatively pseudo-coherent complex along a ring map of finite presentation preserves relative pseudo-coherence. invoked at proof lines 21521-21530. Identifies the tensor product of the complex with the localization of the polynomial algebra with the complex over the localized algebra.

Source reference. Original lemma 064Z: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16283–16283. The reference is applied twice, at proof lines 21516-21518 and at proof lines 21536-21538, to move relative pseudo-coherence to

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the polynomial algebra over the base; the presentations that allow it are inside the counted span.

Source reference. Original lemma 0650: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16345–16345. The reference is applied at proof lines 21521-21527 to identify the tensor product of the complex with the localization of the polynomial algebra with the complex over the localized algebra; the second transfer built on that is inside the counted span.

Lemma 0.2. *Let R be a ring, $f \in R$ an element, $R_f \to A$ is a finite type ring map, $g \in A$, and $K^\bullet$ a complex of A-modules. If $K^\bullet$ is m-pseudo-coherent (resp. pseudo-coherent) relative to R_f, then $K^\bullet \otimes_A A_g$ is m-pseudo-coherent (resp. pseudo-coherent) relative to R.*

Proof. First we show that $K^\bullet$ is m-pseudo-coherent relative to R. Namely, suppose $R_f[x_1, \ldots, x_n] \to A$ is surjective. Write $R_f = R[x_0]/(fx_0 - 1)$. Then $R[x_0, x_1, \ldots, x_n] \to A$ is surjective, and $R_f[x_1, \ldots, x_n]$ is pseudo-coherent as an $R[x_0, \ldots, x_n]$-module. Hence by Lemma 064Z we see that $K^\bullet$ is m-pseudo-coherent as a complex of $R[x_0, x_1, \ldots, x_n]$-modules.

Choose an element $g' \in R[x_0, x_1, \ldots, x_n]$ which maps to $g \in A$. By Lemma 0650 we see that

$$\begin{aligned} K^\bullet \otimes^{\mathbf{L}}_{R[x_0,x_1,\ldots,x_n]} R[x_0, x_1, \ldots, x_n, \frac{1}{g'}] &= K^\bullet \otimes_{R[x_0,x_1,\ldots,x_n]} R[x_0, x_1, \ldots, x_n, \frac{1}{g'}] \\ &= K^\bullet \otimes_A A_f \end{aligned}$$

is m-pseudo-coherent as a complex of $R[x_0, x_1, \ldots, x_n, \frac{1}{g'}]$-modules. write

$$R[x_0, x_1, \ldots, x_n, \frac{1}{g'}] = R[x_0, \ldots, x_n, x_{n+1}]/(x_{n+1}g' - 1).$$

As $R[x_0, x_1, \ldots, x_n, \frac{1}{g'}]$ is pseudo-coherent as a $R[x_0, \ldots, x_n, x_{n+1}]$-module we conclude (see Lemma 064Z) that $K^\bullet \otimes_A A_g$ is m-pseudo-coherent as a complex of $R[x_0, \ldots, x_n, x_{n+1}]$-modules as desired. □